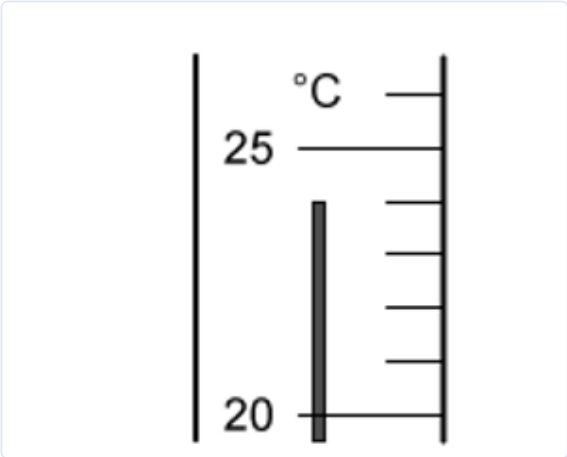
CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

1.3 Errors & Uncertainties

Syllabus Section: 1. Physical Quantities & Units **Paper Type:** Multiple Choice (Paper 1) **Total:** 22 Questions (22 Marks)

Question 1 Easy [1 mark]

The diagram shows part of a thermometer.



What is the correct reading on the thermometer and the uncertainty in this reading?

	Reading/°C	Uncertainty in reading/°C
A	24	±1
B	24	±0.1
C	24.0	±0.2
D	24.0	±0.5

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The uncertainty of a measuring instrument is estimated as plus or minus half the smallest scale division.
- For the thermometer in the question, each mark represents 1.0° C, so the uncertainty is ± 0.5° C
- The reading from this thermometer is 24.0° C, so the result can be written as 24.0 ± 0.5° C

A & B are incorrect because the readings are not quoted to the correct precision, and the uncertainties are incorrect

C is incorrect because even though the precision of the reading is correct, the uncertainty is too low

Question 2 Easy [1 mark]

In an experiment to determine the acceleration of free fall using a falling body, what would lead to a value that is too large?

A air resistance

B dimensions of the body are too large

C measured distance longer than true distance

D measured time longer than true-time

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- To determine the acceleration of free fall, we will need to consider the equation
  - $s = u \quad t + \frac{1}{2}a \quad t^2$
  - We can simplify this to $h = \frac{1}{2}g \quad t^2$ or $g = \frac{2h}{t^2}$ since the falling body will initially be at rest, so u = 0 (h = height, g = acceleration of free fall, t = time)
- If the measured distance, h, was measured to be longer than the true distance, this would increase the measured value of g as the two quantities are directly proportional

A is incorrect because air resistance can be treated as negligible in this experiment, but even if we did factor it in, air resistance would increase the time which would lead to a decreased value of g

B is incorrect because the dimensions of the body are not relevant to the experiment

D is incorrect because if the time were measured to be longer than the true time, this would lead to a decreased value of g since it is inversely proportional to the square of the time (inverse square law)

Question 3 Easy [1 mark]

A metre rule is used to measure the length of a piece of wire. It is found to be 70 cm long to the nearest millimetre.

How should this result be recorded in a table of results?

A 0.7 m

B 0.70 m

C 0.700 m

D 0.7000 m

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- The metre rule can measure to the nearest millimetre, so the answer must be recorded to 3 significant figures
- This is because:
  - A metre rule has a resolution of 1 mm = 0.1 cm = 0.001 m
  - So the measurement of 70 cm must be recorded to the same level of precision = 70.0 cm = 0.700 m

Question 4 Easy [1 mark]

An experiment is done to measure the acceleration of free fall of a body from rest.

Which measurements are needed?

A The height of fall and the time of fall

B The height of fall and the weight of the body

C The mass of the body and the height of fall

D The mass of the body and the time of fall

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- To determine the measurements needed to calculate the acceleration of free fall, we will need to consider the equation
  - $s = u \quad t + \frac{1}{2}a \quad t^2$
  - We can simplify this to $h = \frac{1}{2} \quad g \quad t^2$ since the falling body will initially be at rest, i.e. $u = 0$ (h = height, g = acceleration of free fall, t = time)
- Rearranging the equation above we can see $g = \frac{2h}{t^2}$ so the only measurements required are the height and time of the fall

Question 5 Easy [1 mark]

A micrometer is used to measure the diameters of two cylinders

diameter of first cylinder = 12.78 ± 0.02 mm

diameter of second cylinder = 16.24 ± 0.03 mm

The difference in the diameters is calculated.

What is the uncertainty in this difference?

A ± 0.01 mm

B ± 0.02 mm

C ± 0.03 mm

D ± 0.05 mm

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- If we are adding or subtracting quantities then we simply add the absolute uncertainties together
  - 0.02 + 0.03 = 0.05
- So the uncertainty in the difference in diameters is ± 0.05 mm

Question 6 Easy [1 mark]

A student carries out a series of determinations of the acceleration of free fall g. The table shows the results.

g / m s^{-2}
4.91
4.89
4.88
4.90
4.93
4.92

What can be said about this experiment?

A it is accurate and precise

B it is accurate but not precise

C it is not accurate and not precise

D it is not accurate but is precise

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Precise measurements are consistent, they fluctuate slightly about a mean value
- The measurements are, however, consistent so they are precise
- The true value of g is 9.81 m s^{-2} so the values are not accurate
- The measurements are, however, consistent so they are precise

Question 7 Medium [1 mark]

The measurement of a physical quantity may be subject to random errors and systematic errors.

Which statement is correct?

A random errors can be reduced by taking the average of several measurements

B random errors are always caused by the person taking the measurement

C a systematic error cannot be reduced by adjusting the apparatus

D a systematic error results in a different reading each time the measurement is taken

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- Random errors affect precision, meaning they cause differences in measurements which causes a spread about the mean
- Repeating the measurements at least three times and taking an average of them is one way to reduce random error because it reduces spread about the mean and also enables any anomalies to be spotted and discounted from the results

B is incorrect because random errors can be the result of either the person taking the measurement or the equipment being used

C is incorrect because systematic error affects accuracy and occurs due to the apparatus used or faults in the experimental method so adjusting the apparatus would help reduce systematic error

D is incorrect because systematic error affects accuracy, which causes all results to be too high or too low by the same amount each time

Question 8 Medium [1 mark]

In an experiment, a radio-controlled car takes 2.50 ± 0.05 s to travel 40.0 ± 0.1 m.

What is the car's average speed and the uncertainty in this value?

A 16 ± 1 m s^{-1}

B 16.0 ± 0.2 m s^{-1}

C 16.0 ± 0.4 m s^{-1}

D 16.0 ± 0.36 m s^{-1}

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- $average \quad speed = \frac{distance}{time} = \frac{40.0}{2.50} = 16.0 \quad m \quad s^{-1}$ (3 s.f.)
- To work out uncertainty in data which is multiplied or divided we add the percentage uncertainties
  - $\% \quad uncertainty = \frac{uncertainty}{value} \times 100$
  - $\frac{0.05}{2.50} \times 100 + \frac{0.1}{40} \times 100 = \quad 2.0 \quad \% \quad + \quad 0.25 \quad \% \quad = \quad 2.25 \quad \%$
- Then we find 2.25 % of 16 which is $16 \times \frac{2.25}{100} = 0.36$
- The precision of the uncertainty must match the precision of the value so average speed = 16.0 ± 0.4 m s^{-1}

Question 9 Medium [1 mark]

Four students each made a series of measurements of the acceleration of free fall *g*. The table shows the results obtained.

Which set of results could be described as precise but not accurate?

	g / ms^{-2}			
A	9.81	9.79	9.84	9.83
B	9.81	10.12	9.89	9.94
C	9.45	9.21	8.99	8.76
D	8.45	8.46	8.50	8.41

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- An accurate measurement is one which is close to the true value
- Precise measurements are consistent and they fluctuate slightly about a mean value
- The true value of *g* is 9.81 m s^{-2} so the values given in **D** are not accurate
- The measurements in **D** are, however, consistent so they can be described as precise

A is incorrect because these measurements can be described as accurate and precise

B is incorrect because these measurements can be described as accurate but not precise

C is incorrect because these measurements are neither accurate or precise

Question 10 Medium [1 mark]

Which experimental technique reduces the systematic error of the quantity being investigated?

A adjusting an ammeter to remove its zero error before measuring a current

B measuring several internodal distances on a standing wave to find the mean internodal distance

C measuring the diameter of a wire repeatedly and calculating the average

D timing a large number of oscillations to find a period

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- Systematic error affects accuracy and occurs due to the apparatus used or faults in the experimental method so adjusting the apparatus would help reduce systematic error

B, C & D are incorrect because these all involve repeating measurements and taking an average which is a way to reduce random error but not systematic error

Question 11 Medium [1 mark]

A steel rule can be read to the nearest millimetre. It is used to measure the length of a bar whose true length is 895 mm. Repeated measurements give the following readings.

length / mm	892, 891, 892, 891, 891, 892

Are the readings accurate and precise to within 1 mm?

	Results are accurate to within 1 mm	Results are precise to within 1 mm
A	no	no
B	no	yes
C	yes	no
D	yes	yes

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- An accurate measurement is one which is close to the true value
- Precise measurements are consistent and they fluctuate slightly about a mean value
- The mean value of the measurements is 891.5 mm and the true value is 895 mm so the values are not accurate within 1mm
- The measurements are, however, consistent within 1 mm so they can be described as precise

Question 12 Medium [1 mark]

The Young modulus of the material of a wire is to be found. The Young modulus E is given by the equation below.

$$E = \frac{4Fl}{\pi d^2 x}$$

The wire is extended by a known force and the following measurements are made.

Which measurement has the largest effect on the uncertainty in the value of the calculated Young modulus?

	Measurement	Symbol	Value
A	length of wire before force applied	l	2.043 ± 0.002 m
B	diameter of wire	d	0.54 ± 0.02 mm
C	force applied	F	19.62 ± 0.01 N
D	extension of wire with force applied	x	5.2 ± 0.2 mm

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **B** because:*

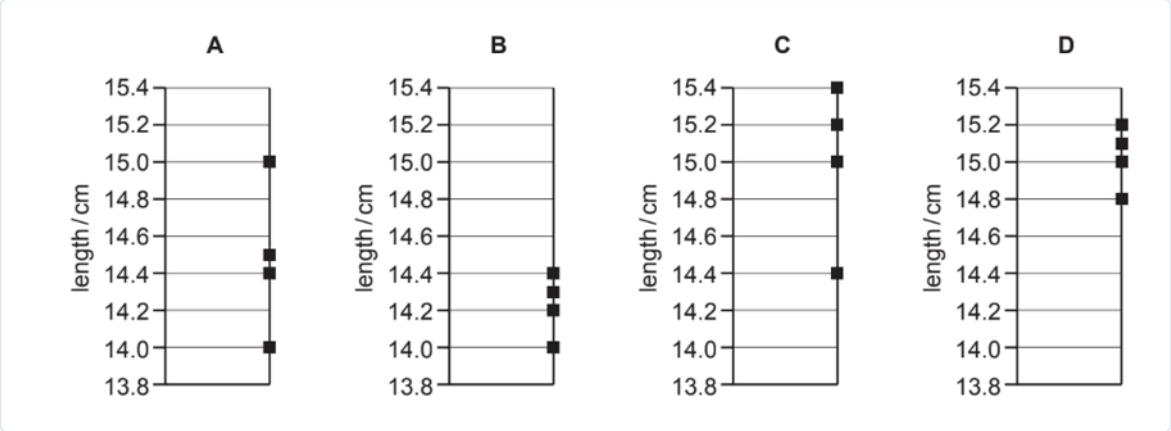
- To find out which value has the greatest effect on the overall uncertainty, we have to compare the percentage uncertainty of each value
  - length, l : $\frac{0.002}{2.043} \times 100 = 0.1 \quad \%$
  - diameter, d^2 : $2 \times \frac{0.02}{0.54} \times 100 = 7.4 \quad \%$
  - force, F : $\frac{0.01}{19.62} \times 100 = 0.05 \quad \%$
  - extension, x : $\frac{0.2}{5.2} \times 100 = 3.8 \quad \%$
- The value with the largest percentage uncertainty is diameter

Exam Tip: The diameter has the largest effect on the value of Young Modulus due it being squared. When a quantity is raised to a power, you must multiply the percentage uncertainty by the power

Question 13 Medium [1 mark]

Four different students use a ruler to measure the length of a 15.0 cm pencil. Their measurements are recorded on four different charts.

Which chart shows measurements that are precise but **not** accurate?



A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- An accurate measurement is one which is close to the true value
- Precise measurements are consistent and they fluctuate slightly about a mean value
- The values fall outside of the 14.8 - 15.2 cm range which would be considered accurate, so the measurements in **B** can be described as not accurate
- The measurements are, however, consistent within the 14.0 - 14.4 cm range so they can be described as precise

A is incorrect because these measurements are not precise or accurate

C is incorrect because three of the measurements can be described as accurate and precise, but the outlier makes it neither of those

D is incorrect because these measurements are both precise and accurate

Question 14 Medium [1 mark]

What is the ratio $\frac{10^2\ \text{THz}}{10^{-4}\ \text{pHz}}$?

A 10^{-30}

B 10^{-2}

C 10^{2}

D 10^{30}

Mark Scheme & Explanation:

The correct answer is D

- $T = \text{tera} = 10^{12}$

$$10^{2} \times 10^{12} = 10^{(12+2)} = 10^{14}$$

- $p = \text{pico} = 10^{-12}$

$$10^{-4} \times 10^{-12} = 10^{(-12-4)} = 10^{-16}$$

- Determine the ratio

$$\frac{10^{14}}{10^{-16}} = 10^{14--16} = 10^{30}$$

- This is answer **D**

Question 15 Hard [1 mark]

A student uses a digital ammeter to measure a current. The reading of the ammeter is found to fluctuate between 1.98 A and 2.02 A.

The manufacturer of the ammeter states that any reading has a systematic uncertainty of ±1%.

Which value of current should be quoted by the student?

A (2.00 ± 0.01) A

B (2.00 ± 0.02) A

C (2.00 ± 0.03) A

D (2.00 ± 0.04) A

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The mean value of the current is $\frac{1.98+2.02}{2} = 2.00$ A
- The fluctuation between 1.98 A and 2.02 A is caused by random errors, so the random uncertainty is ±0.02 A
- The systematic uncertainty is ±1% which is 2.00 × $\frac{1}{100}$ = 0.02 A
- The total uncertainty = random uncertainty + systematic uncertainty
  - 0.02 + 0.02 = ±0.04 A
- So the current quoted should be 2.00 ± 0.04 A

Question 16 Hard [1 mark]

The uncertainty in the value of the momentum of a trolley passing between two points X and Y varies with the choice of measuring devices.

Measurements for the same trolley made by different instruments were recorded.

1	distance between X and Y using a metre rule with cm divisions = 0.55 m
2	distance between X and Y using a metre rule with mm divisions = 0.547 m
3	timings using a wristwatch measuring to the nearest 0.5 s at X = 0.0 s and at Y = 4.5 s
4	timings using light gates measuring to the nearest 0.1 s at X = 0.0s and at Y = 4.3 s
5	mass of trolley using a balance measuring to the nearest g = 6.4×10^{-2} kg
6	mass of trolley using a balance measuring to the nearest 10g = 6×10^{-2} kg

Which measurements, one for each quantity measured, leads to the least uncertainty in the value of the momentum of the trolley?

A 1, 3 and 6

B 1, 4 and 6

C 2, 3 and 6

D 2, 4 and 5

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Looking at the equation for momentum
  - $p = mv = \frac{ms}{t}$
  - $\frac{\Delta p}{p} = \frac{\Delta m}{m} + \frac{\Delta s}{s} + \frac{\Delta t}{t}$
- We can see that for the uncertainty in momentum (Δ p) to have the least value, the uncertainties in all the quantities should be as small as possible

- The instruments which measure to a higher degree of precision will lead to less uncertainty in the value of momentum
    - The metre rule in 2 has a greater precision than 1 because the smallest scale division is in mm which is more accurate than cm.
    - The light gate in 4 has a greater precision than the wristwatch in 3 because 4.3 s is more accurate than 4.5 s
    - The balance which measures to the nearest g has a greater precision than the balance which measures to the nearest 10 g as it measures to an extra decimal point making the measurement more accurate

Question 17 Hard [1 mark]

In an experiment to determine the acceleration of free fall *g*, a ball bearing is held by an electromagnet. When the current to the electromagnet is switched off, a clock starts and the ball bearing falls. After falling a distance *h*, the ball bearing strikes a switch to stop the clock which measures the time *t* of the fall.

If systematic errors cause *t* and *h* to be measured incorrectly, which error **must** cause *g* to appear greater than 9.81 m s^{-2}?

A *h* measured as being **smaller** than it actually is and *t* is measured correctly

B *h* measured as being **smaller** than it actually is and *t* measured as being **larger** than it actually is

C *h* measured as being **larger** than it actually is and *t* measured as being **larger** than it actually is

D *h* is measured correctly and *t* measured as being **smaller** than it actually is

Mark Scheme & Explanation:

*The correct answer is **D** because:*

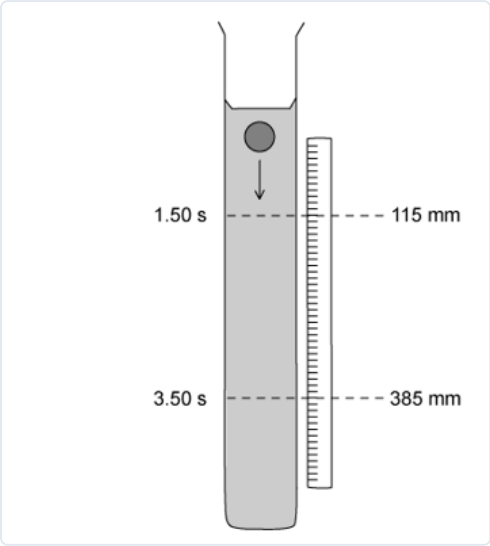
- To determine the relationship between *g, h* and *t* we will need to consider the equation
  - $s = ut + \frac{1}{2}a \quad t^2$
- Since the falling body will initially be at rest, $u = 0$, the equation simplifies to
  - $h = \frac{1}{2}g \quad t^2$
- Rearranging the equation above we can see
  - $g = \frac{2h}{t^2}$
- We can see that *g* and *h* are directly proportional so as *h* increases, *g* increases and vice versa
- We can see that *g* and *t* are inversely proportional, so as *t* increases, *g* decreases and vice versa
- Therefore, if *h* is measured correctly, *g* will be calculated correctly but if *t* is measured to be lower than it should this will cause *g* to be larger than it should be

A is incorrect because this would lead to a value of g which is smaller than it should be

B & C are incorrect because without knowing how much larger or smaller these quantities are, it's not possible to say if *g* will be larger or smaller

Question 18 Hard [1 mark]

The diagram shows an experiment to measure the speed of a small ball falling at constant speed through a clear liquid in a glass tube.



There are two marks on the tube. The top mark is positioned at 115 ± 1 mm on the adjacent rule and the lower mark at 385 ± 1 mm. The ball passes the top mark at 1.50 ± 0.02 s and passes the lower mark at 3.50 ± 0.02 s.

The constant speed of the ball is calculated to be 135 mm s^{-1}

Which expression calculates the fractional uncertainty in the value of this speed?

A $\frac{2}{270} + \frac{0.04}{2.00}$

B $\frac{1}{270} + \frac{0.02}{2.00}$

C $\frac{1}{115} + \frac{0.02}{1.50}$

D $\frac{1}{385} + \frac{0.02}{3.50}$

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The equation relating speed, distance and time is given by
  - $speed \quad v = \frac{distance \ \ s}{time \ \ t}$
- The value and uncertainty in distance is:
  - s = 385 – 115 = 270 mm

- Δ s = ± (1 + 1) = ± 2 mm

- The value and uncertainty in time is:
  - t = 3.50 – 1.50 = 2.00 s
  - Δ t = ± (0.02 + 0.02) = ± 0.04 s

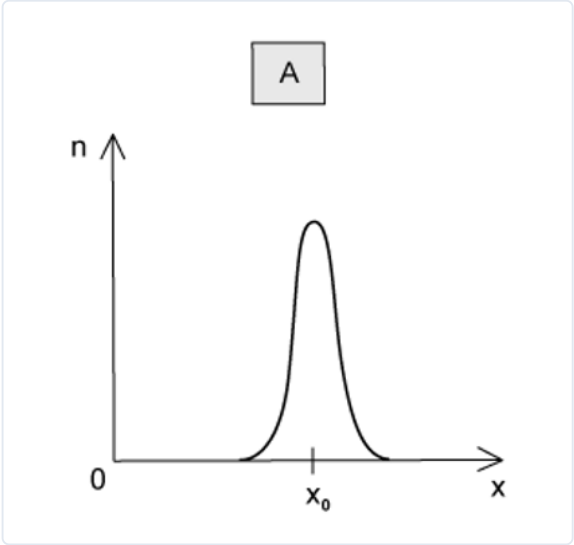
- To work out the fractional uncertainty in speed:
  - $\frac{\Delta v}{v} = \frac{\Delta s}{s} + \frac{\Delta t}{t}$
  - $\frac{\Delta v}{v} = \frac{2}{270} + \frac{0.04}{2.00}$

Question 19 Hard [1 mark]

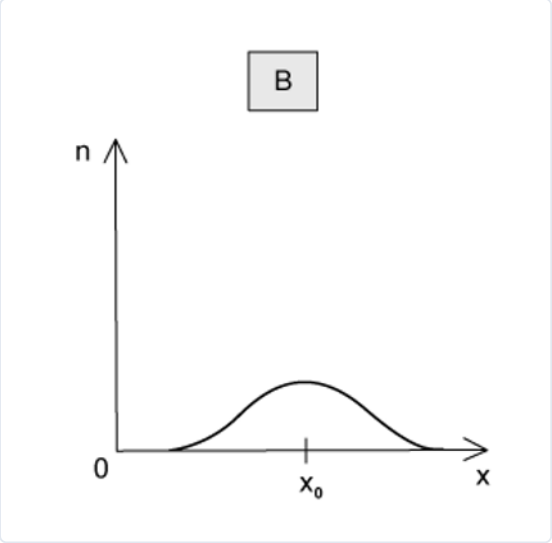
A fixed quantity x_0 is measured many times in an experiment that has experimental uncertainty. A graph is plotted to show the number n of times that a particular value x is obtained.

Which graph could be obtained if the measurement of x_0 has a large systematic error but a small random error?

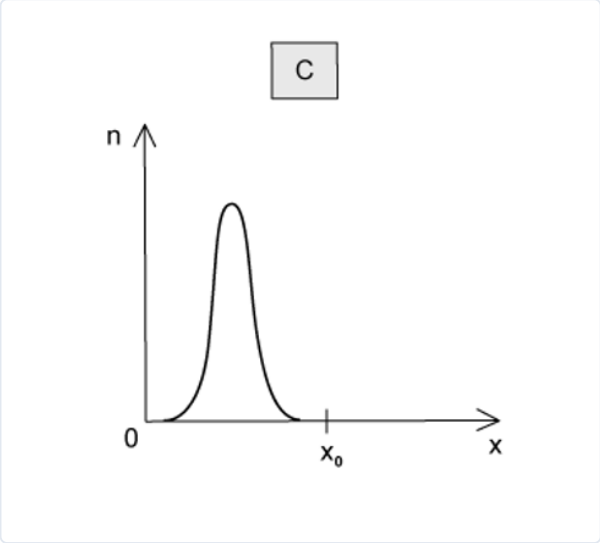
A



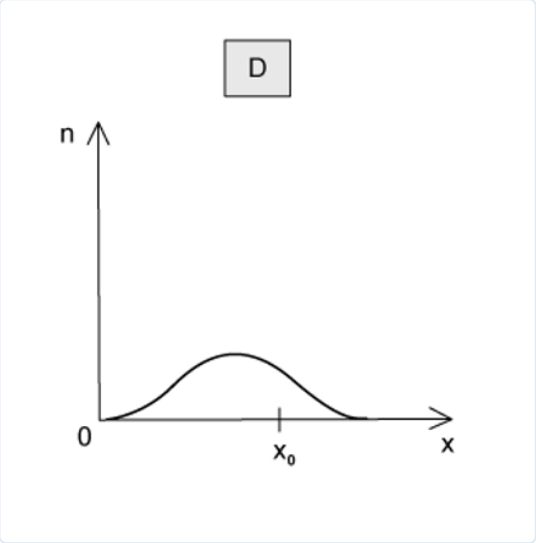
B



C



D



Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Random errors affect precision, meaning they cause differences in measurements which causes a spread about the mean. Graph **C** shows a narrow curve, with a high peak so the measurements are close to each other, meaning the random error is low
- Systematic error affects accuracy, which causes all results to be too high or too low by the same amount each time. Graph **C** shows the measurements are consistent but they are much lower meaning the systematic error is high

A is incorrect because this graph shows a set of results with both low random and systematic errors

B is incorrect because this graph shows a set of results which have a low systematic error and a high random error

D is incorrect because this graph shows a set of results with both high random and systematic errors

Question 20 Hard [1 mark]

A student finds the density of a liquid by measuring its mass and volume. The following is a summary of his measurements.

mass of empty beaker = (20 ± 1) g

mass of beaker + liquid = (70 ± 1) g

volume of liquid = (10.0 ± 0.5) cm^3

He correctly calculates the density of the liquid as 5.0 g cm^{-3}.

What is the uncertainty in this value?

A 0.3 g cm^{-3}

B 0.5 g cm^{-3}

C 0.6 g cm^{-3}

D 2.6 g cm^{-3}

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- The equation relating density, mass and volume is given by
  - $density \quad \rho = \frac{mass m}{volume v}$
- The value and uncertainty in mass is:
  - m = 70 – 20 = 50 g
  - Δm = ± (1 + 1) = ± 2 g
- The value and uncertainty in volume is:
  - V = 10 cm^3
  - ΔV = ± 0.5 cm^3
- To work out the uncertainty in density:
  - $\frac{\Delta \rho}{\rho} = \frac{\Delta m}{m} + \frac{\Delta V}{V}$
  - $\frac{\Delta \rho}{\rho} = \frac{2}{50} + \frac{0.5}{10} = 0.04 + 0.05 = 0.09$
  - $\frac{\Delta \rho}{\rho} = 0.09, where \quad \rho = 5.0 \quad g \quad cm^{-3} \quad$ so

- $\Delta\rho = 5 \times 0.09 = 0.45 = 0.5 \quad \mathrm{g} \quad \mathrm{cm}^{-3} 2 \quad s.f$

Question 21 Hard [1 mark]

A steel wire is stretched in an experiment to determine the Young modulus for steel.

The uncertainties in the measurements are given below.

Measurement	Uncertainty
load on wire	±2%
length of wire	±0.2%
diameter of wire	±1.5%
extension	±1%

What is the percentage uncertainty in the Young modulus?

A 1.3%

B 1.8%

C 4.7%

D 6.2%

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The equation for Young Modulus is:
  - $Young \quad Modulus \quad = \quad \frac{ForceF}{AreaA} \times \frac{lengthl}{extensionx} = \frac{load \times length}{\frac{\pi}{4} \times diameter^2 \times extension}$
- To find the percentage uncertainty in Young Modulus we must add the uncertainties in the measurements
- We can ignore the $\frac{\pi}{4}$ since it is a constant with no units, and it is important to multiply the uncertainty in the diameter by 2 since the diameter is squared
  - % uncertainty = 2 + 0.2 + (2 × 1.5) + 1 = 6.2%

Question 22 Hard [1 mark]

A mass is dropped from rest and falls through a distance of 2.0 m in a vacuum. An observer records the time taken for the mass to fall through this distance using a manually operated stopwatch and repeats the measurements a further two times. The average result of these measured times, displayed in the table below, was used to determine a value for the acceleration of free fall. This was calculated to be 9.8 m s^{-2}.

	First measurement	Second measurement	Third measurement	Average
Time/ s	0.6	0.73	0.59	0.64

Which statement best relates to the experiment?

A The measurements are precise and accurate with no evidence of random errors.

B The measurements are not accurate and not always recorded to the degree of precision of the measuring device but the calculated experimental result is accurate.

C The measurements are not always recorded to the degree of precision of the measuring device but are accurate. Systematic errors may be present.

D The range of results shows that there were random errors made but the calculated value is correct so the experiment was successful.

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Precision is how close the measured values are to each other
- In a measuring instrument, the precision is indicated by the number of significant figures given by the instrument
- From the measurements, we can see that the readings are not very close to each other. This scatter of readings indicates the presence of random errors
- Even though the same instrument is used, the measurements are not given to the same number of decimal places, so the measurements are not precise
- Accuracy is how close a measured value is to the true value
- We can see that the individual measurements are not close to the average value of time. So, the measurements are not accurate, but the calculated experimental is correct

A is incorrect because even though the correct experimental value was achieved, the results do not show accuracy or precision so the experiment was not successful

C is incorrect because systematic error affects accuracy and occurs due to the apparatus used or faults in the experimental method, so if systematic errors are present then the measurements are not accurate

D is incorrect because the significant random errors indicate that the experiment was not successful, even though the calculated value is close to the actual value